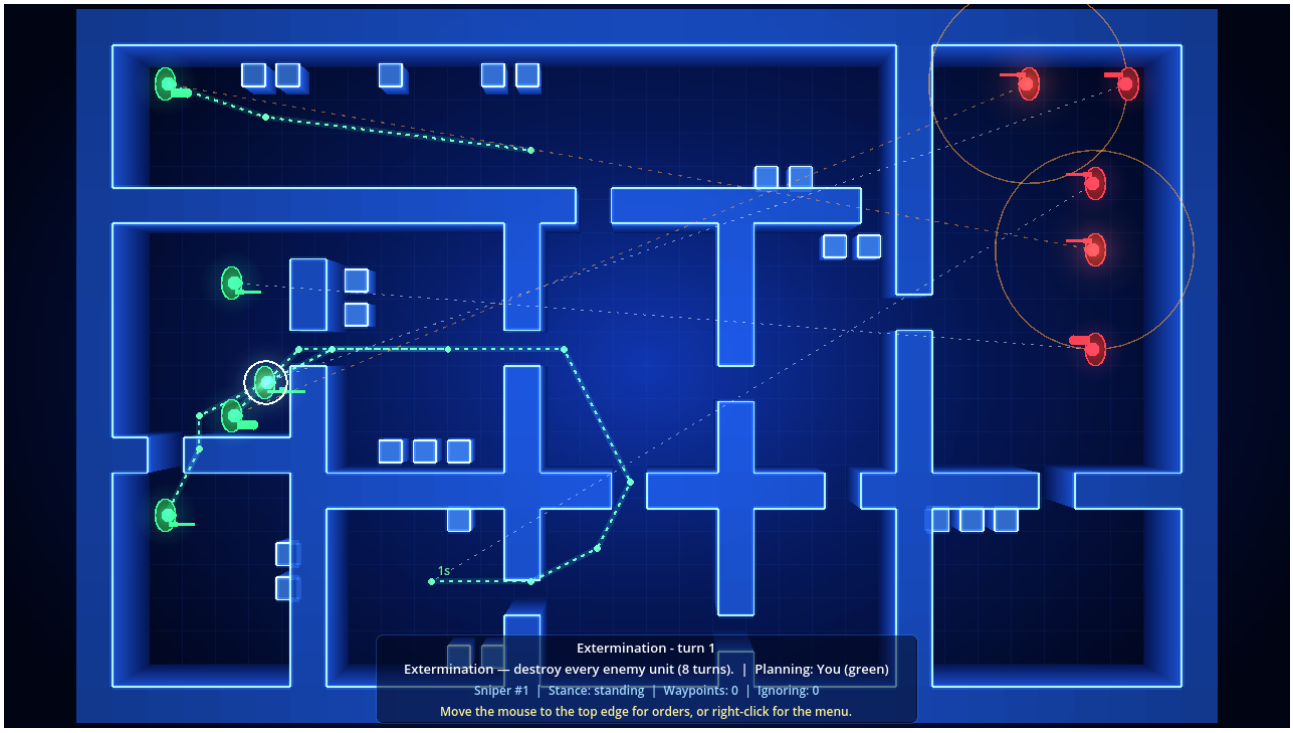
COLD SYNAPSE

A personal-use, installable rebuild of a simultaneous-turn tactics game. Project overview and play-test preview.



Executive summary. The game is playable end to end: five original game modes plus the Red-style Upload mode, Light and Dark play, versus a computer opponent or two people on one screen, and a campaign of 55 missions in 19 chapters plus two extra packs (12 missions each). Combat follows the original's deterministic time-to-kill rules. What is missing is packaging (an installable Linux app) and balance tuning from real play. The look follows the original: a glowing, extruded 3D-style neon map with cube half-walls, figure silhouettes, white tracers and bloom explosions, drawn procedurally. The orders interface stays hidden until you need it. Everything below was captured from the running game, not mocked up.

Built so far	Status
Step 1 Deterministic simulation core	Done, tested
Step 2 Map generator, game modes, match flow	Done, tested
Step 3 Planning UI, Test preview, replay, hotseat, computer opponent	Done, tested
Step 3.5 Rules rebuilt to the original (time-to-kill, real mode rules, Upload)	Done, tested
Step 4 Campaign: 55 missions, Red-style pack, Blackout pack, mission engine	Done, tested
Step 5 Linux packaging (AppImage and .deb)	Not started - your call

Automated checks: 166 simulation, mode, AI, UI-flow and campaign assertions pass, and all 79 missions load and run two AI turns. Nobody has play-tested for balance or feel yet - that is what this document is meant to help you decide.

1. How it plays

Both sides plan the same five seconds at the same time. Nothing is random: identical orders always produce identical results, so the Test preview shows you exactly what your own plan would do against whatever you guess the enemy will do.



Orders and controls

Action	How
Orders interface	Hidden by default. Move the mouse to the top edge to slide the orders bar in, or right-click for a context menu.
Select a unit / add a waypoint	Click a unit, then click the map. Paths route around walls automatically.
Undo / clear	Backspace removes the last step; Delete clears the unit's orders (also in the right-click menu).
Aim (A)	Click a point. The unit faces it. A unit that is already aiming wins the first shot.
Duck / stand (D), Wait (W)	Duck behind half-height walls to be unseen; wait pauses at the last waypoint.
Fire (F)	Rocket and grenade launchers: click the target point.
Ignore (I), Continue on sight (C)	Skip a specific enemy, or stop engaging entirely to break out of a fight.
Test preview (Space), Commit (Enter)	Preview on a copy of the game; commit locks the turn in.
Replay	Space pauses, +/- changes speed, drag the timeline, Enter skips.

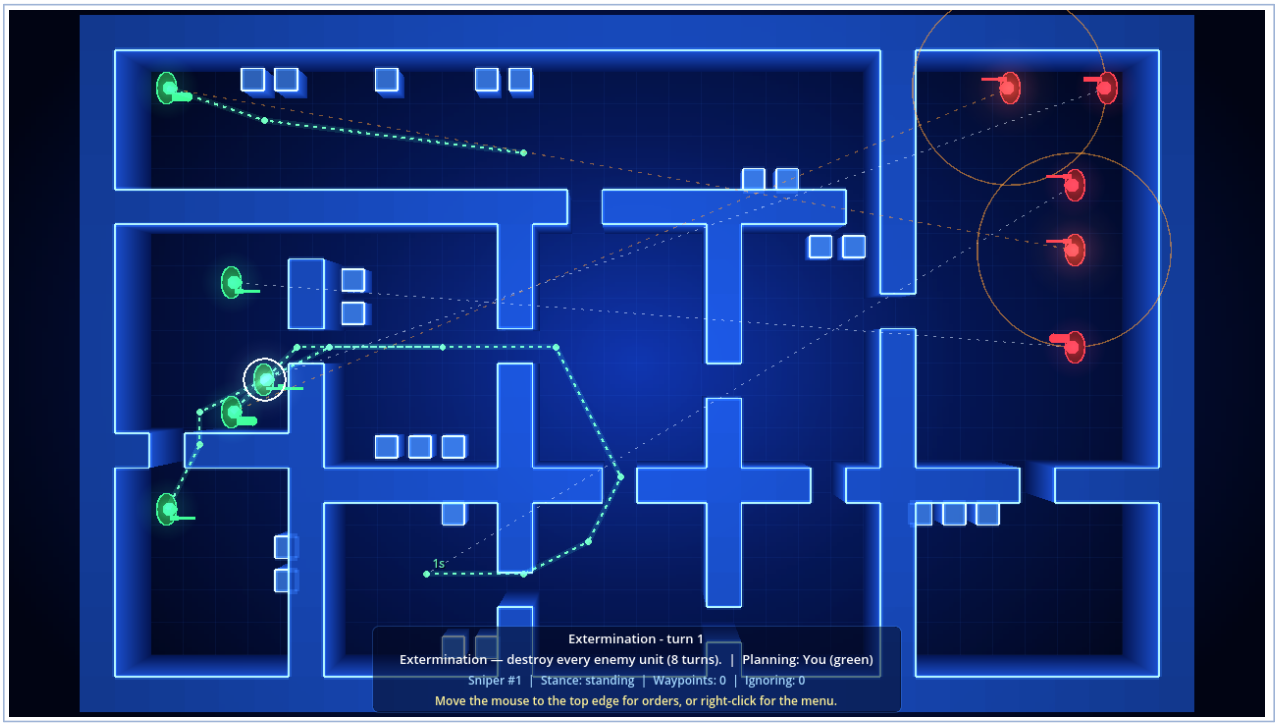
What decides a gunfight

- **Time to kill**, not dice: unit type, distance, cover, how still the shooter is (standing still beats walking while aiming beats running), who saw whom first, and turn delay.
- **Cover:** a ducked unit behind a half wall cannot be seen or shot, but also cannot see out. A standing unit behind one gets a time-to-kill bonus.
- **Weapons:** shotguns win close, machine guns win the middle, snipers win at long range. Rockets destroy walls; grenades bounce and detonate three seconds after launch.
- **Dark mode:** units only see in a roughly 120 degree cone, and enemies show only while seen (an outline marks the last known position).

2. The main screens

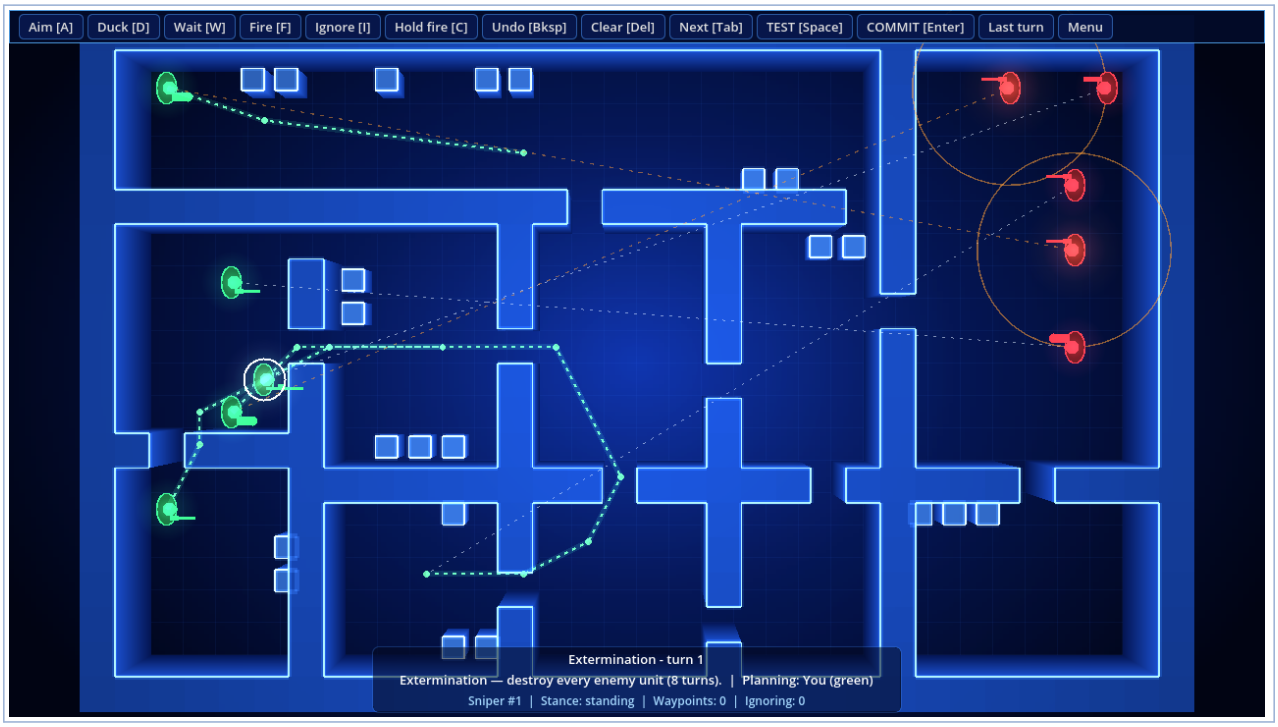


Main menu. Choose a mode, Light or Dark, versus the computer or hotseat, the map seed and the squad size - or open the campaign and packs.

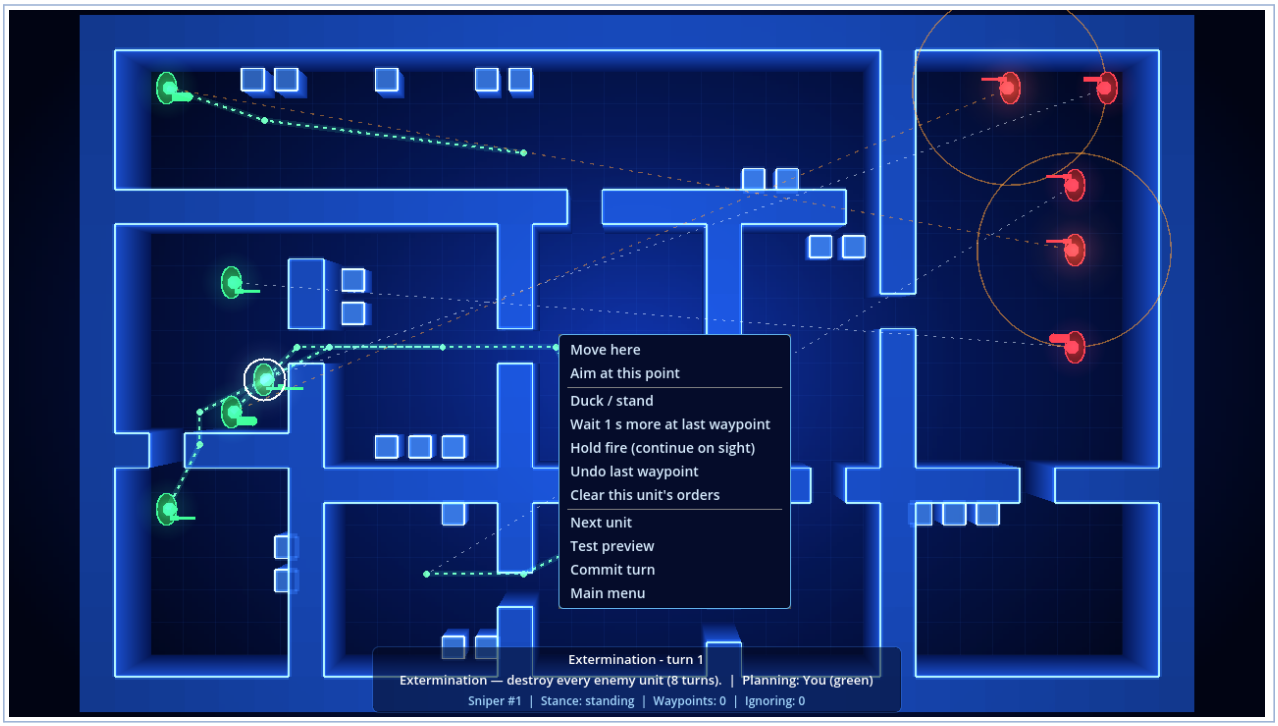


Planning. Green paths and dots are waypoints; dashed lines are aim directions; the orange circles are the blast radius of the two launchers' fire orders; '1s' is a wait. The board fills the whole window; only a slim status line sits at the bottom.

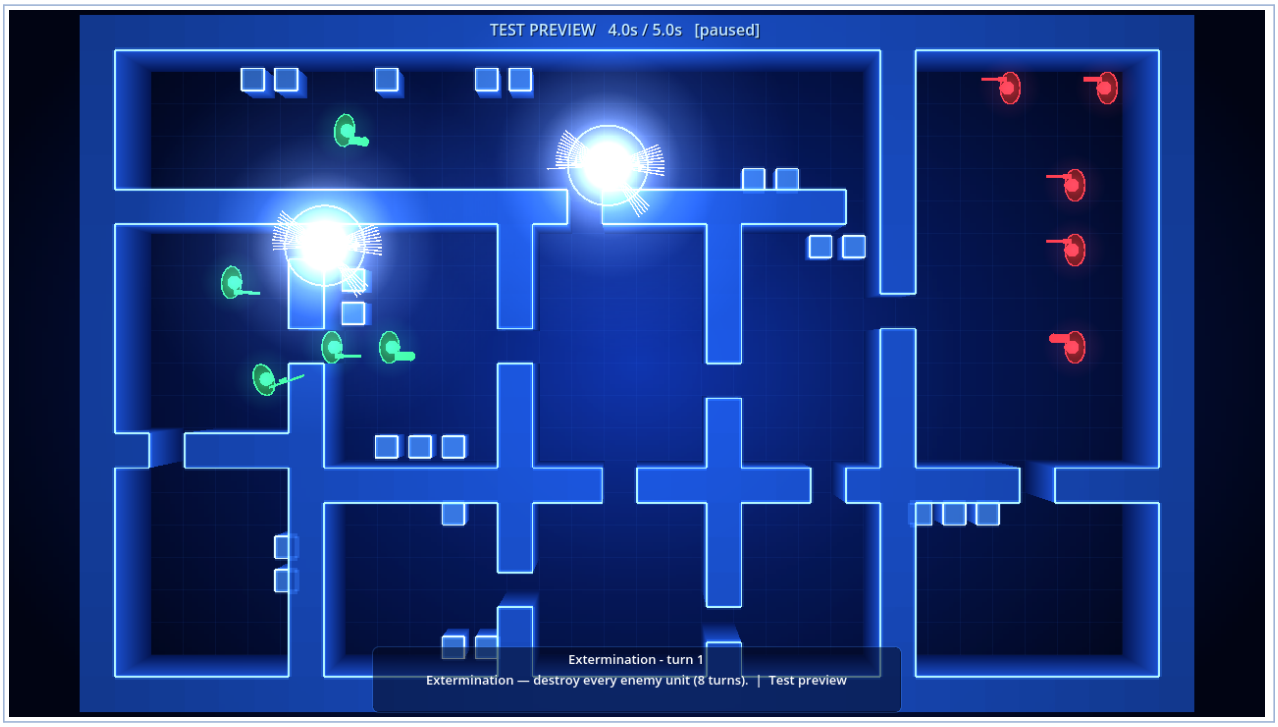
Orders interface and look



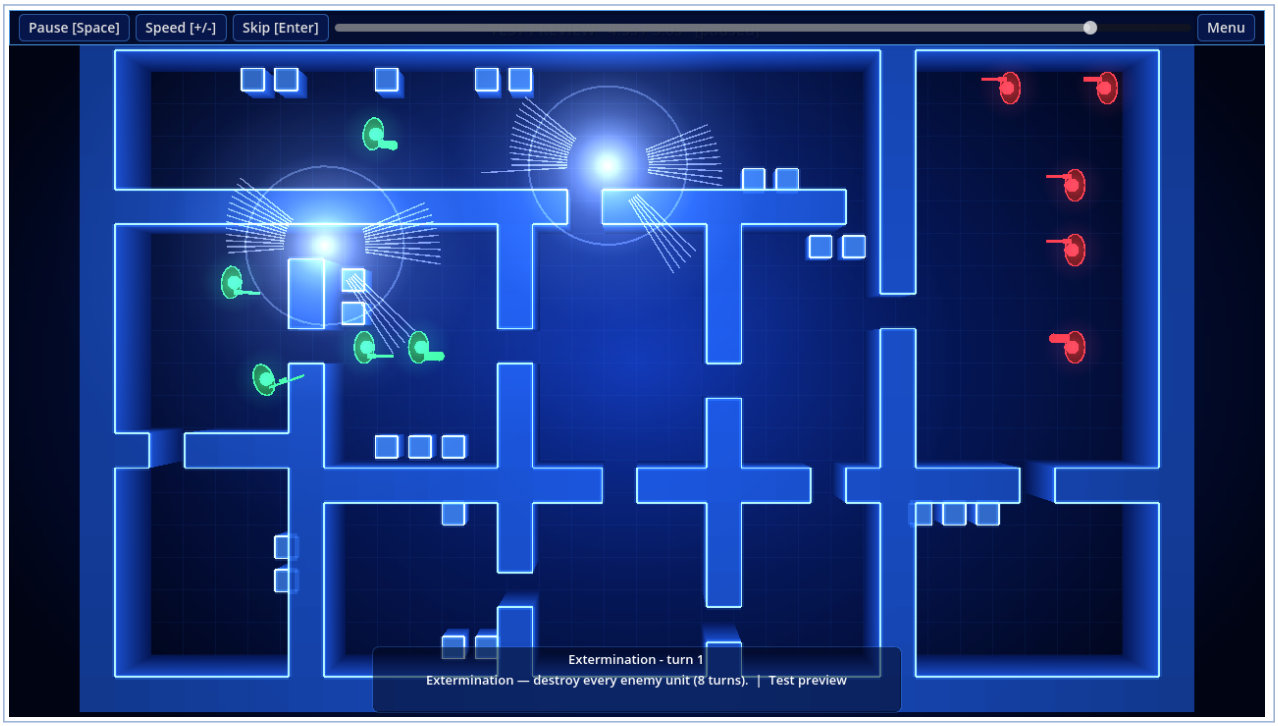
Orders bar. Hidden until the mouse touches the top edge of the window; then it slides in with Aim, Duck, Wait, Fire, Ignore, Hold fire, Undo, Clear, Next, TEST and COMMIT.



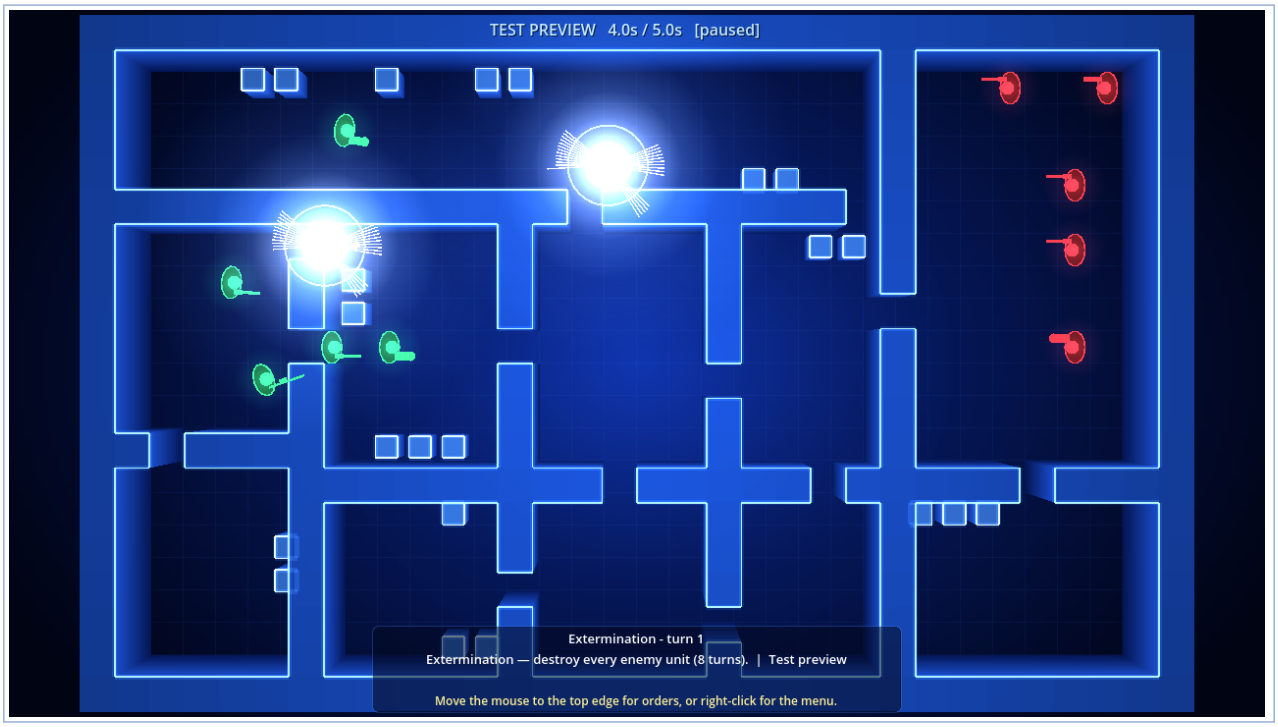
Right-click menu. The same orders at the cursor, plus 'move here', 'aim at this point' and 'ignore this enemy' acting on the exact spot you clicked.



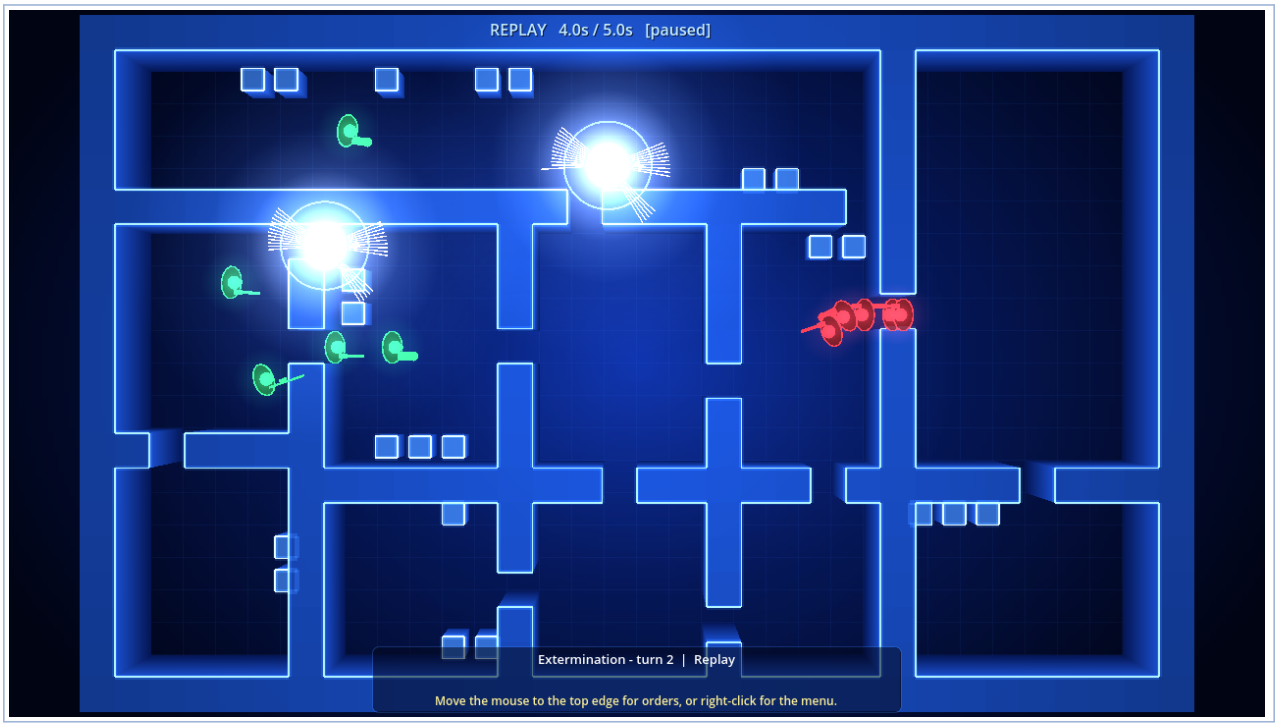
Bloom and sparks. Explosions are white-hot with a wide blue-white glow and radial sparks. Walls are extruded blocks whose faces catch the light; half walls are cubes.



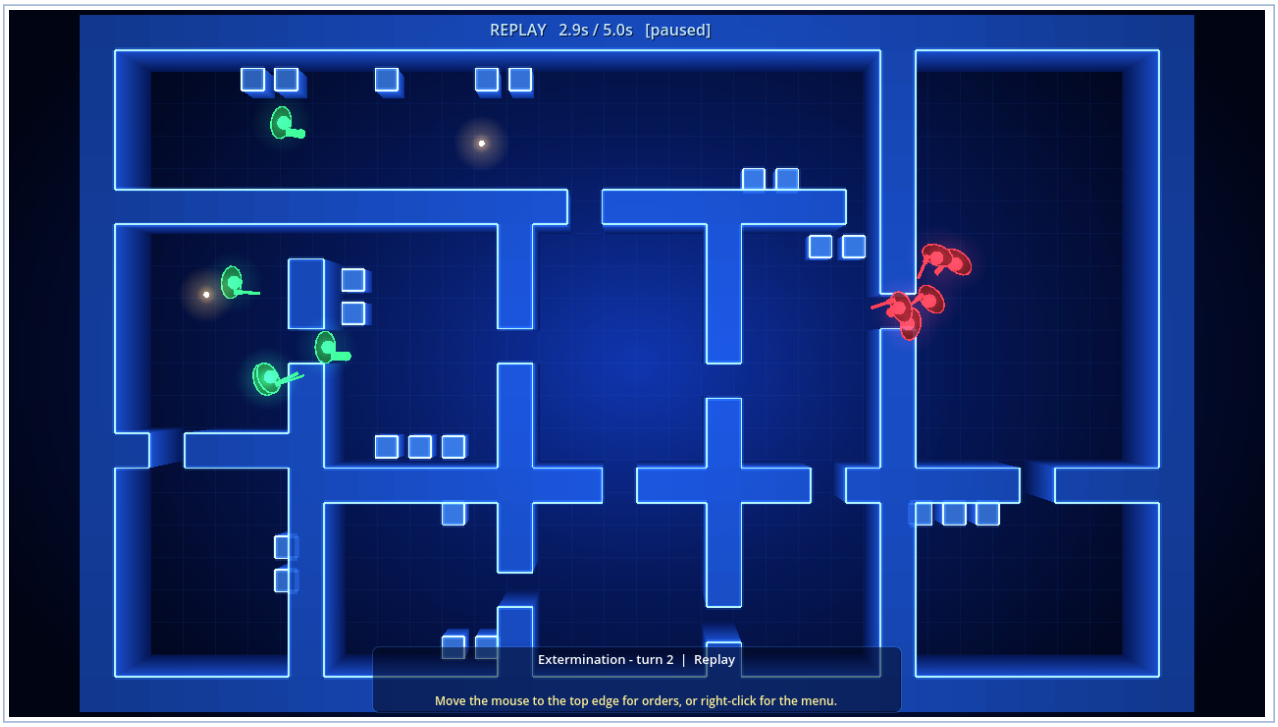
Replay controls. During a replay or preview the top bar shows Pause, Speed, Skip and a draggable timeline.



Test preview. Plays your plan against a copy of the game (the enemy stands still unless you give it hypothetical orders). The real game is untouched. The glowing white bursts are grenade blasts mid-detonation.



Replay of a real turn after committing, with the computer's moves included. Its controls are in the top bar.



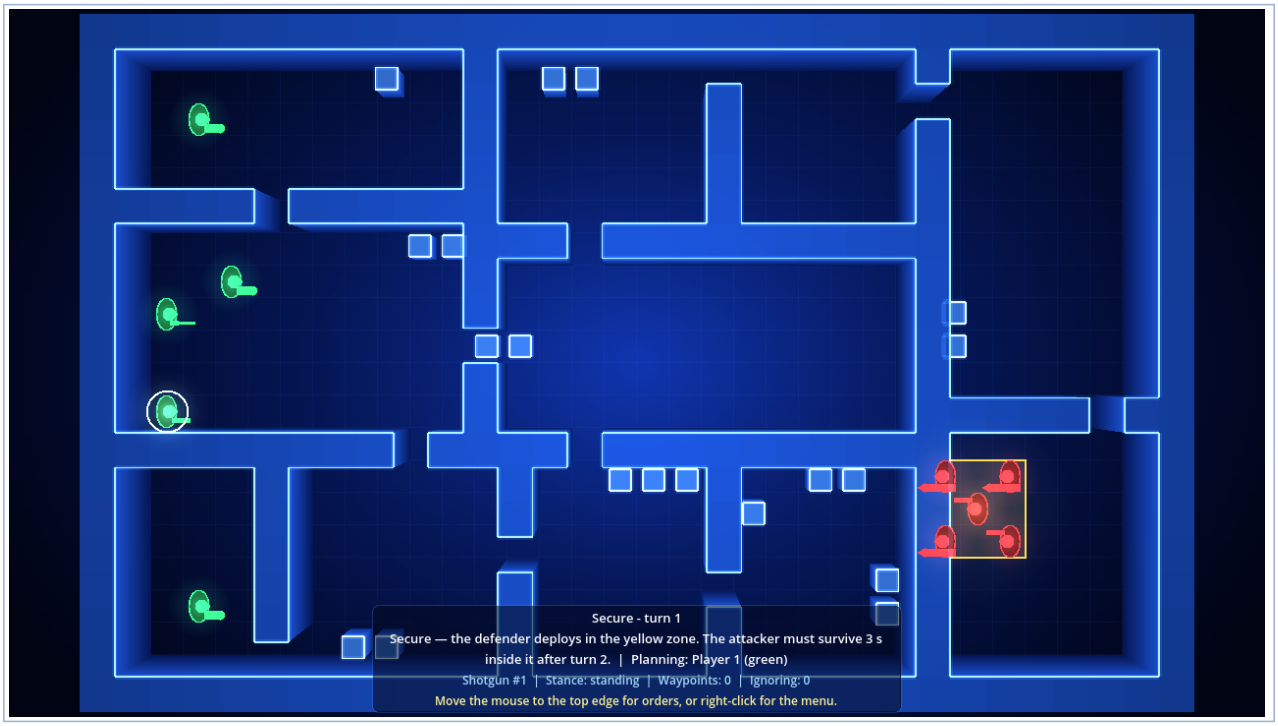
Kill shots appear as short red lines; crosses mark the fallen. Rockets and grenades are drawn in flight.



Dark mode, Disputed. Only the cones your units can see are lit. Outlined red circles are enemies' known starting positions. The green edge strips become extraction zones after turn 4, when the boxes appear.

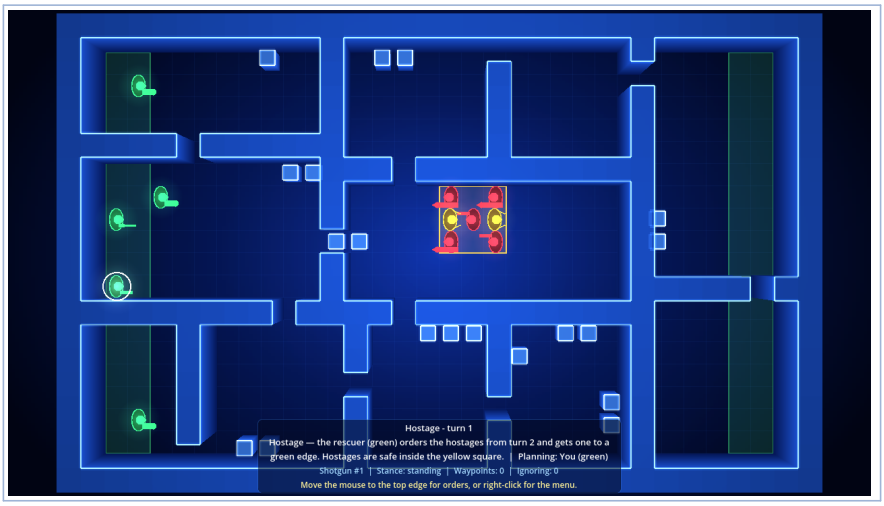


Hotseat. A hand-over screen hides each player's orders from the other. Orders are only applied when both have committed.

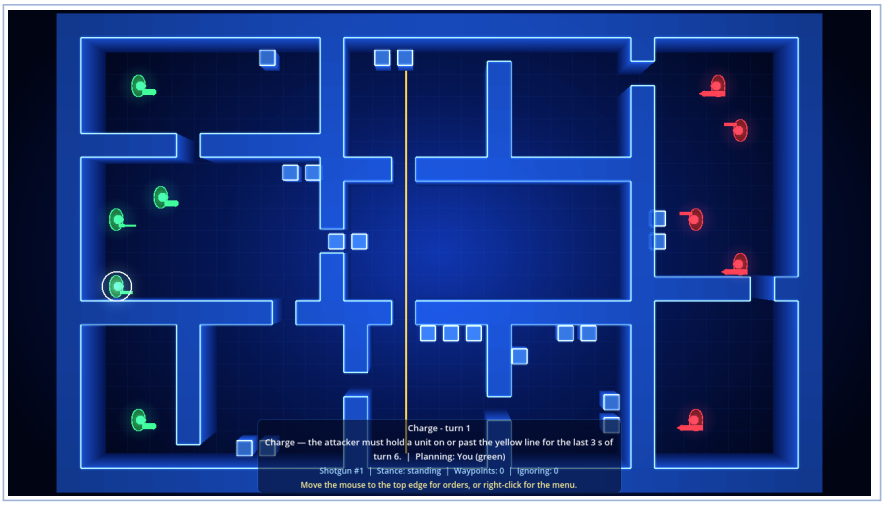


Secure. The defender (right) deploys inside the yellow zone; the attacker must hold it for 3 seconds after turn 2.

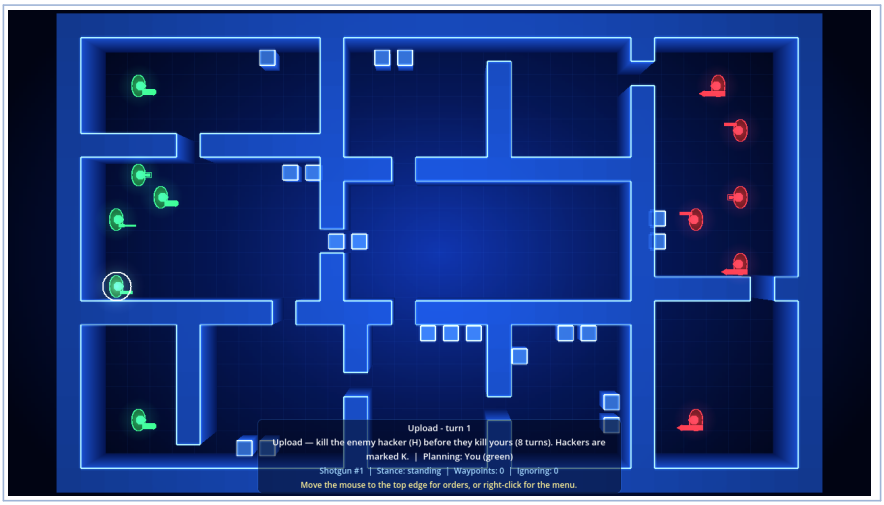
Other modes at a glance



Hostage. Two hostages (yellow H) start in a safe square at the centre. The rescuer can order them from turn 2 and must get one to a green edge.



Charge. The higher bidder attacks and must hold a unit past the yellow line for the last 3 seconds of turn 6.



Upload (Red-style). Each side has a hacker (K); kill theirs before they kill yours. My own design, since the reference material did not cover it.

3. Units and modes

Units

Unit	Role	Notes
Machine Gun	All-rounder	Long range, slows a lot with distance; best when still and pre-aimed.
Shotgun	Close range	Fast mover; shortest time-to-kill up close; useless beyond ~6 tiles.
Sniper	Long range	Very slow to fire, but little affected by range; weak up close.
Rocket Launcher	Indirect	Flies in a straight line, explodes on the first full wall, destroys terrain; a ducked launcher also hits half walls.
Grenade Launcher	Indirect	Lobbed, bounces off walls, detonates 3 s after launch; does not destroy walls.
Riot Shield (Red)	Defensive	No weapon; bulletproof frontal shield while standing still; bounces grenades.
Commando, Devastator, Hacker, civilians	Campaign / Upload	Very fast shotgun; very fast launcher firing several rockets a turn; the Upload target; unarmed neutrals.

Game modes

Mode	Turns	How you win
Extermination	8	Destroy the enemy; otherwise higher score (10 per kill, +20 wipe-out).
Secure	6 Light / 12 Dark	Attacker survives 3 s in the zone after turn 2; defender wins by lasting out.
Disputed	16	Boxes appear after turn 4; carry them into an edge exit. 30 points per box.
Hostage	8	Rescuer gets a hostage to an edge; captors win by killing them.
Charge	6	Attacker holds a unit past the bid line for the last 3 s of turn 6.
Upload (Red-style)	8	Kill the enemy hacker.

4. Campaign and packs

The campaign follows the original's structure: 55 missions in 19 chapters, usually three per location. Names are lightly altered and all briefing text is my own. Missions are plain JSON files you can edit; maps are generated from a fixed seed per mission, so they are repeatable but are not the original hand-made maps.

#	Chapter	Missions
1	Nashar Ward	1 Incursion, 2 Slaughter, 3 Extraction
2	Power Station	4 Incursion, 5 The Breach, 6 Assassination
3	The Broadcast Studio	7 Incursion, 8 Clearance, 9 Broadcast
4	Blue Daylight	10 Defend the Barracks, 11 Attack the Chapel, 12 Projector Takedown
5	Cairn Dwelling	13 Defend the Dwelling, 14 Push Back, 15 Shape Purge
6	Novena Quarter	16 Defend the Base, 17 Counter, 18 The Meeting
7	Ninstar Tower	19 The Lobby, 20 Second Floor, 21 The Corridor
8	Dillenger's Villa	22 Break the Compound, 23 Security Takedown, 24 The Negotiation
9	Site 77	25 Gain Access, 26 Data Harvest, 27 Set Up the Bomb
10	The Devastator	28 First Encounter, 29 Steal the Plans, 30 Facility Takedown
11	Brightmoor	31 Daylight Raid, 32 Hostage Situation, 33 Emmanuele's Doubt
12	Brightmoor Core	34 Connection, 35 Stop the Charge, 36 The Shape Portal
13	Brightmoor Finale	37 The Library, 38 Disrupt the Form, 39 Fighting in the Square
14	Termon Bank	40 Hacking In, 41 Break the Vault, 42 Cashing In
15	Open War	43 Clemens Square, 44 Rapid Takedown, 45 Protection
16	Finding Nox	46 Save Belaqua, 47 Reaching, 48 The Escape
17	Glass Spire	49 Incursion, 50 Courtyard, 51 Access Control
18	Final Hour	52 The Corridor, 53 Ouroboros, 54 The Reversal
19	The Mandala	55 Mandala

Red Pack

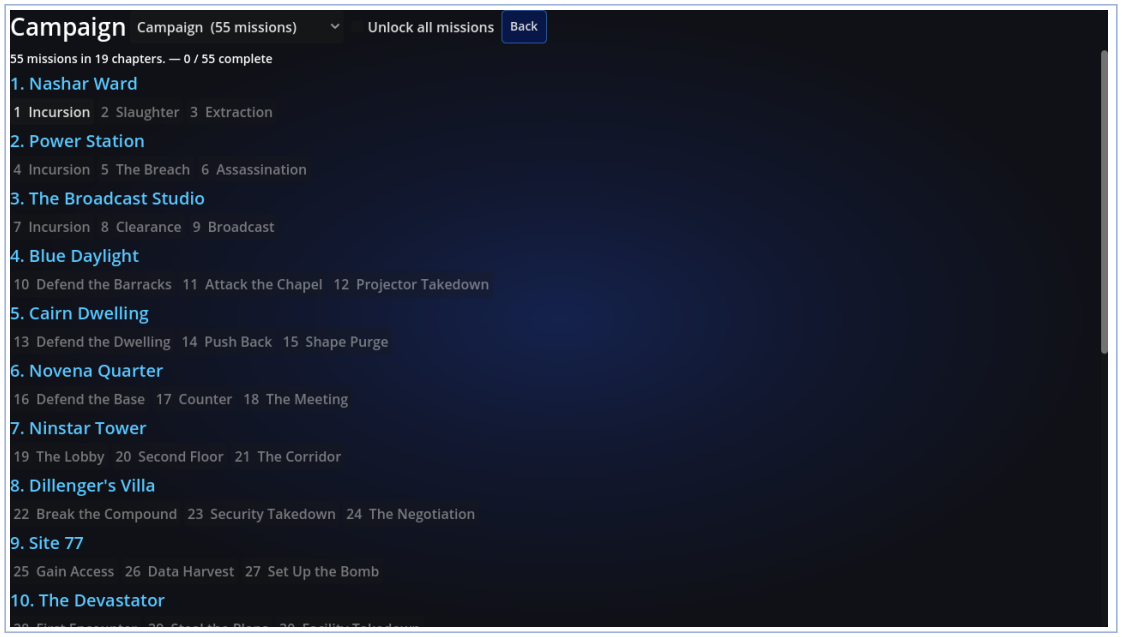
Riot Shield and Upload missions in the style of the Red expansion. Chapters: Red Line (Shield Wall, Cover Fire, Breakwater); Upload Protocol (Handshake, Packet Loss, Root Access); Red Rescue (Escort Duty, Crossfire Street, Glass House); Red Dawn (Siege, Breakout, Redline Final).

Blackout Pack

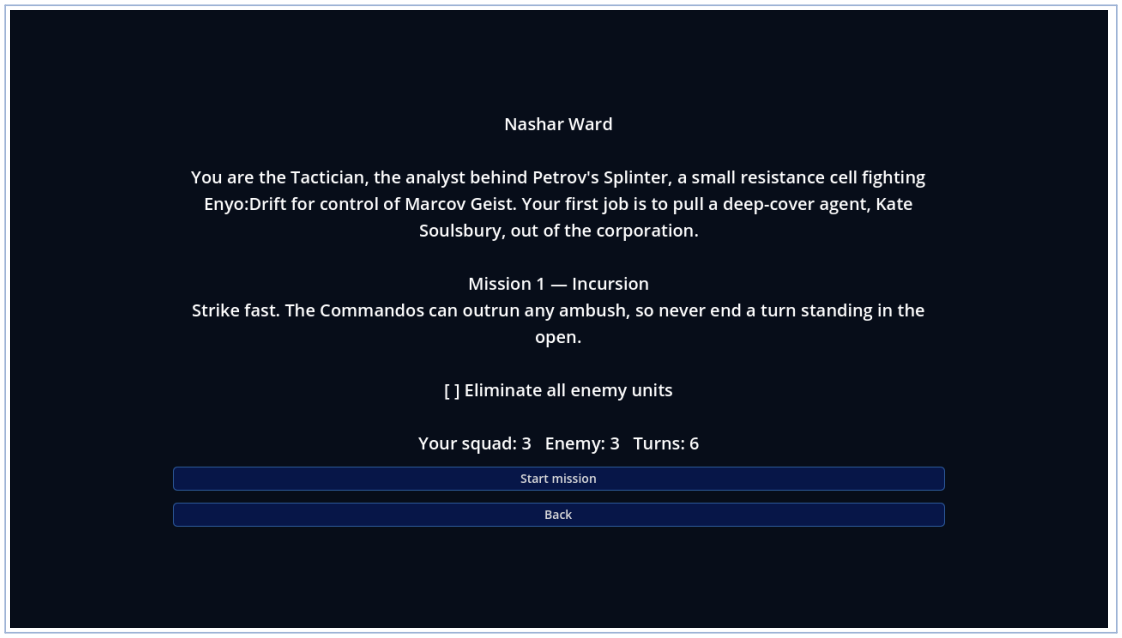
Twelve original missions: night ops, heavy weapons, snipers and a final stand. Chapters: Night Shift (Lights Out, Dead Drop, Skeleton Crew); Iron Rain (Rubble, Bombardment, Hold the Rubble); Quiet Hands (The Long Gallery, Overwatch, Silent Entry); Last Call (Closing Time, The Big Score, Last Call).

Mission objectives supported

Objective	Used for
Eliminate (optionally 'all but N')	Straight fights, e.g. Incursion, The Library, Clemens Square
Kill the marked target	Assassination, Security Takedown, Emmanuele's Doubt
Reach a zone with a given unit	Escorts and infiltrations: Extraction, The Meeting, The Corridor
Collect boxes and carry them to an edge	Broadcast, Shape Purge, Data Harvest, Cashing In
Rescue hostages	Hostage Situation, Save Belaqua
Hold a zone for N turns / survive N turns	The Breach, Stop the Charge, Protection (one 50-second turn)

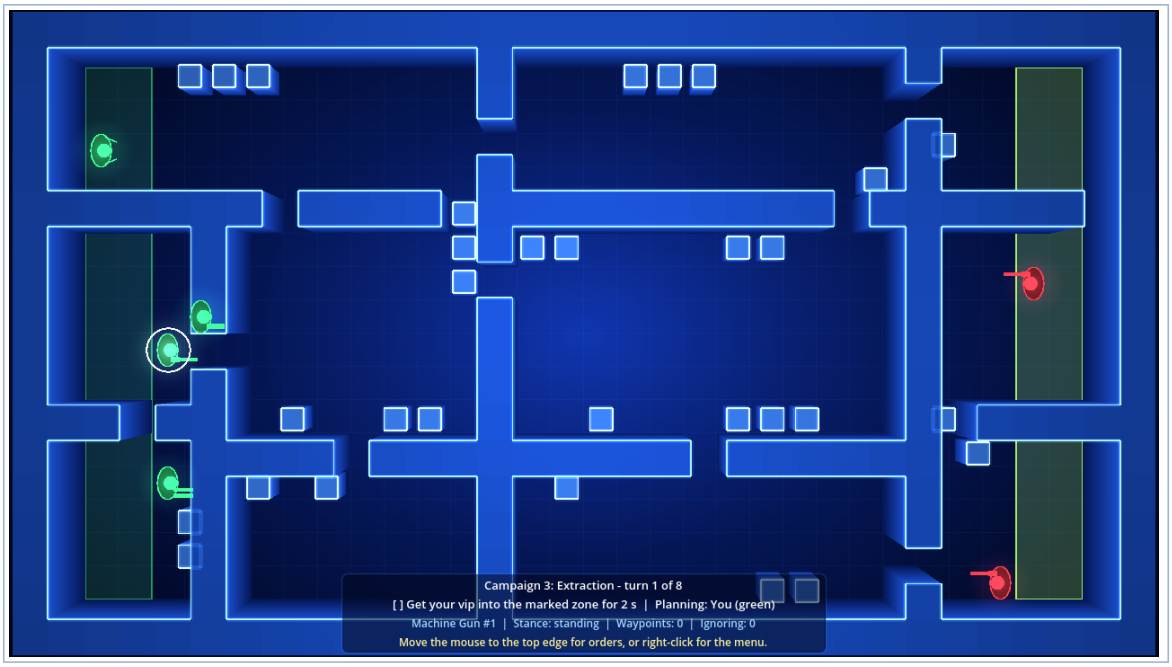


Campaign screen. Chapters with their mission buttons; each mission unlocks when the previous one is done (or use 'Unlock all'). Packs are chosen at the top.

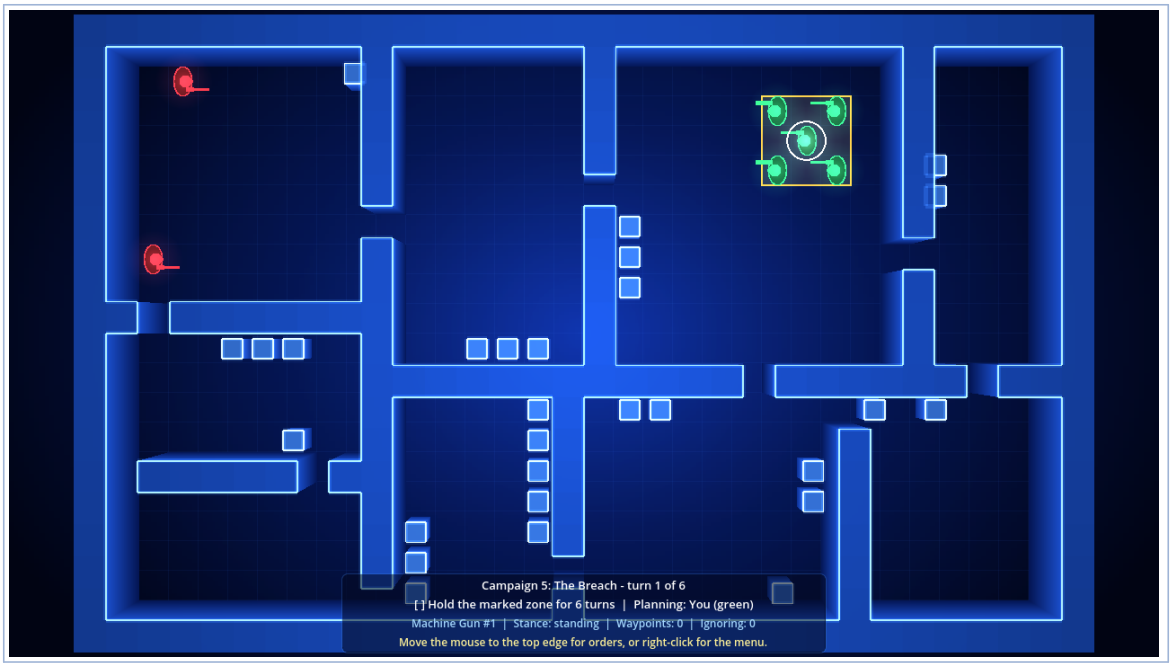


Mission briefing with the chapter story, objectives, squad sizes and turn limit.

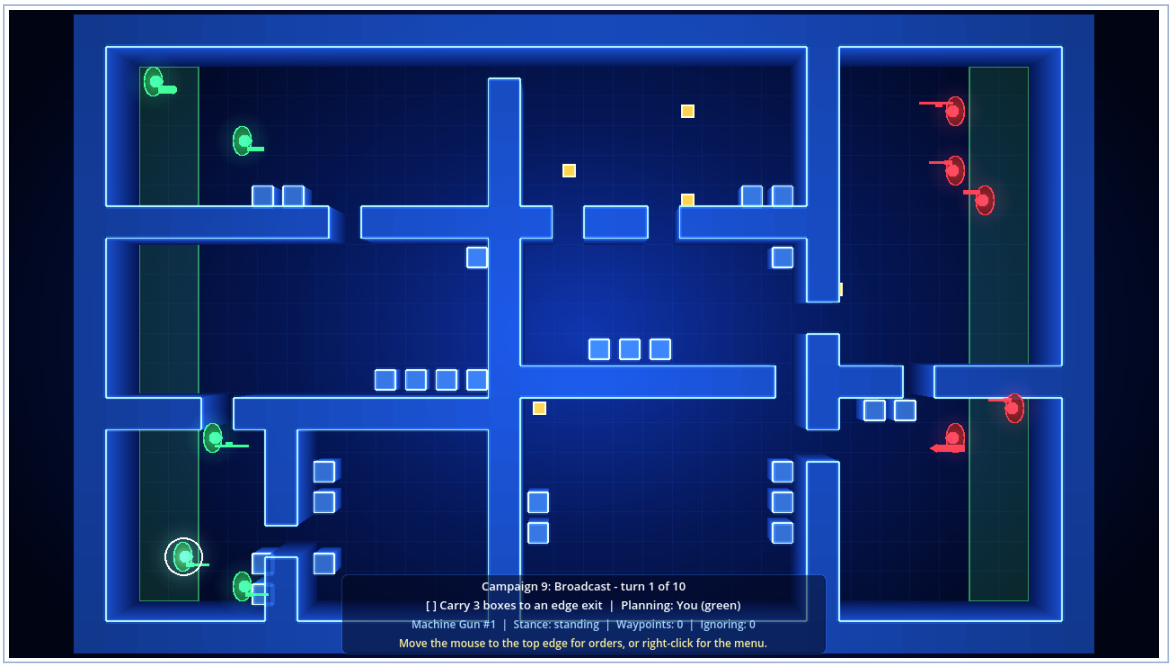
5. Example missions



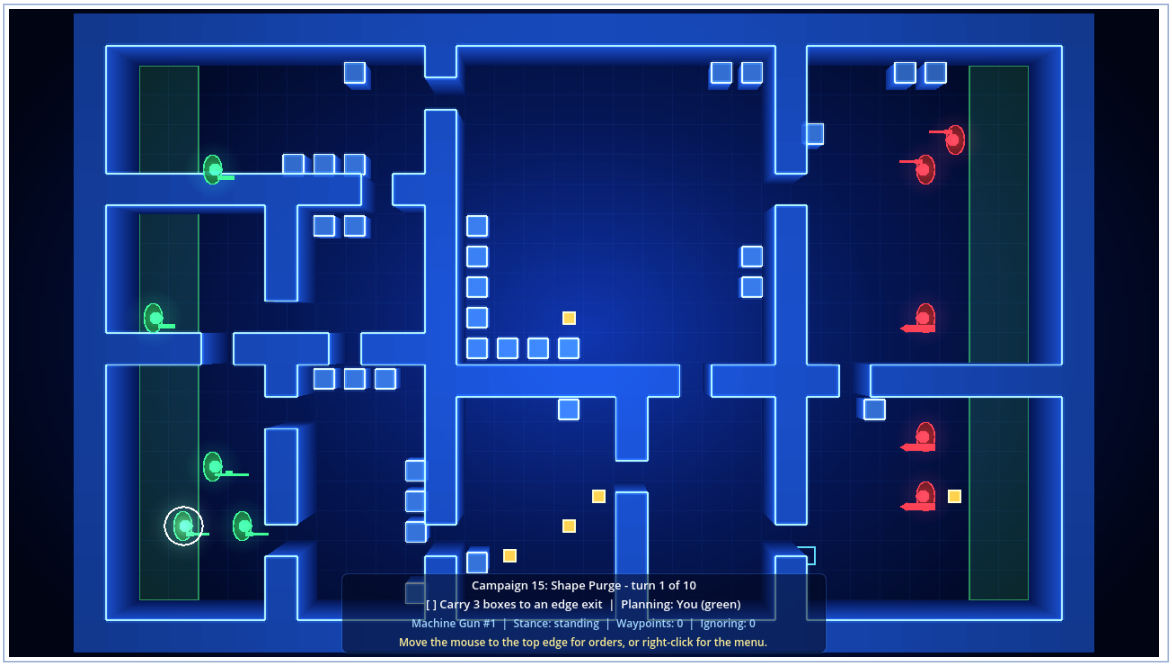
Mission 3 - Extraction. Escort the VIP (the civilian) to the east edge while reinforcements arrive on a timer.



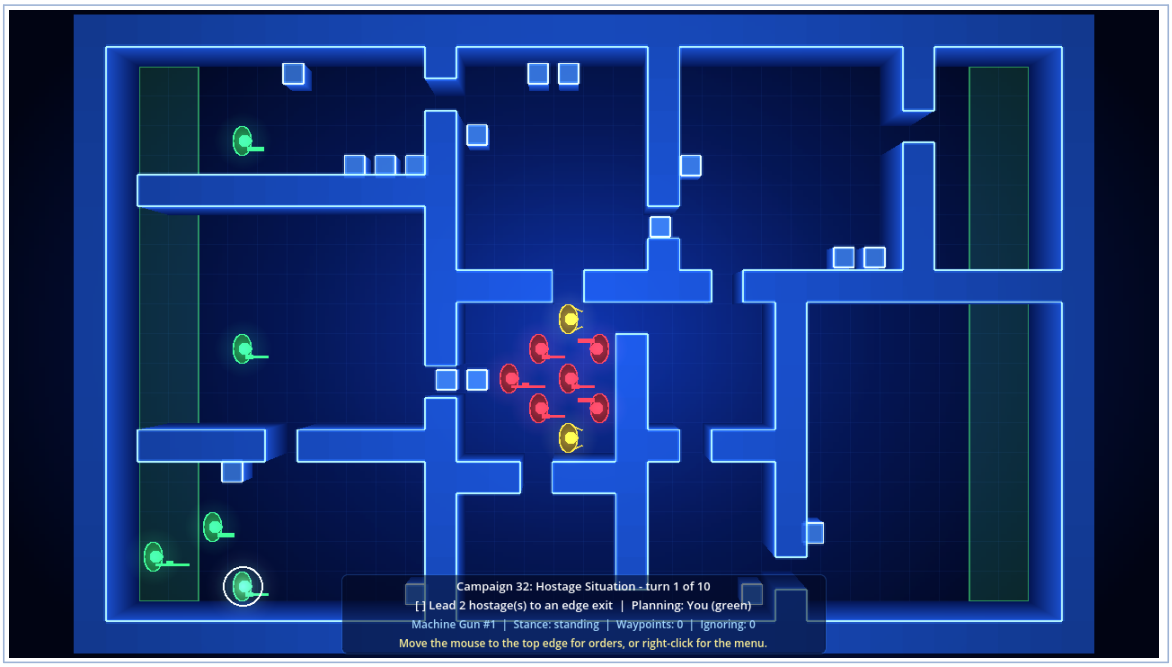
Mission 5 - The Breach. Hold the yellow room for 6 turns with two computer-controlled allies against waves of reinforcements.



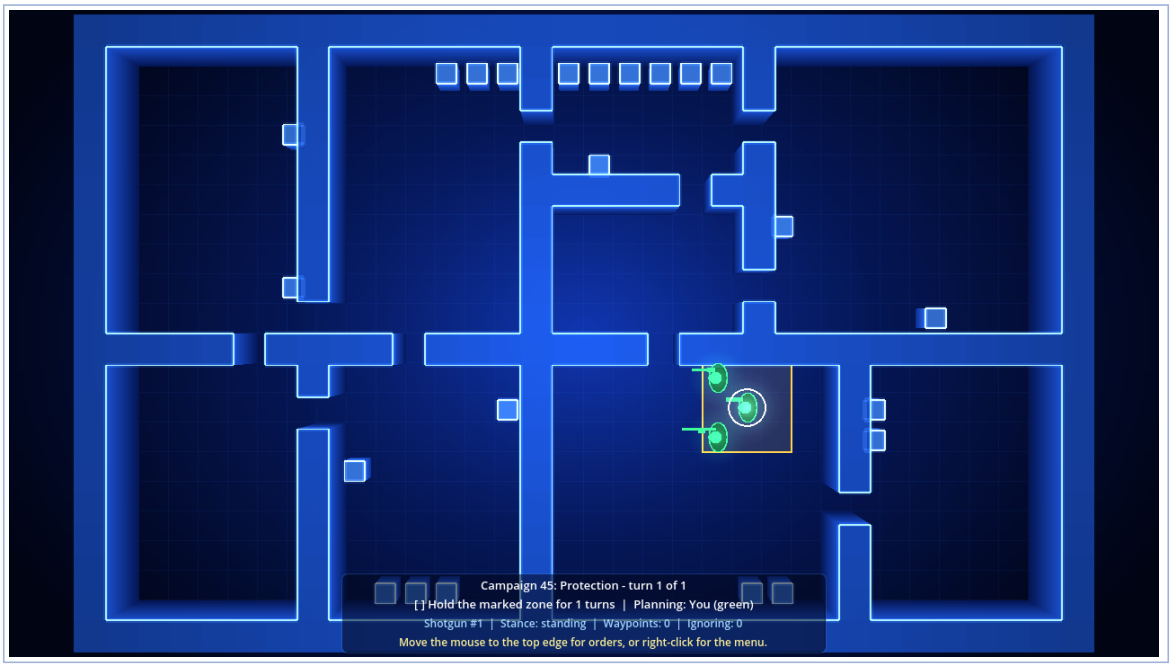
Mission 9 - Broadcast. Five key crates (yellow); carry three to the green edge strips. Enemy rocket launcher, sniper and shotgun in the east.



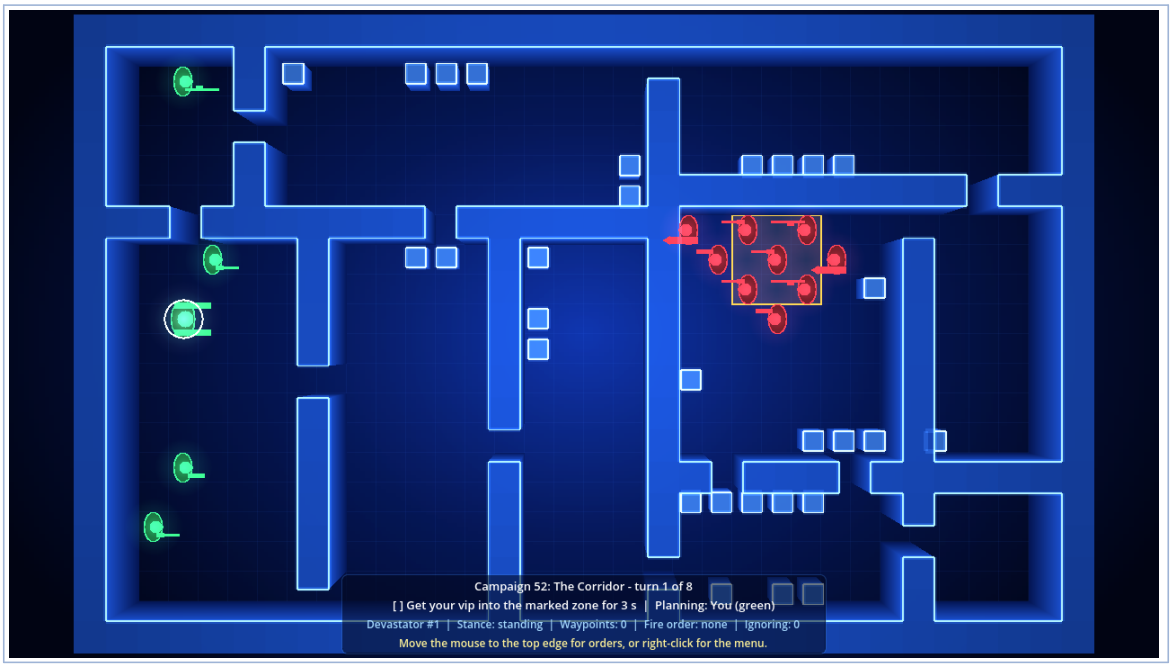
Mission 15 - Shape Purge. Collect three data crates while three rocket launchers pound the map; a hidden panel destroys them.



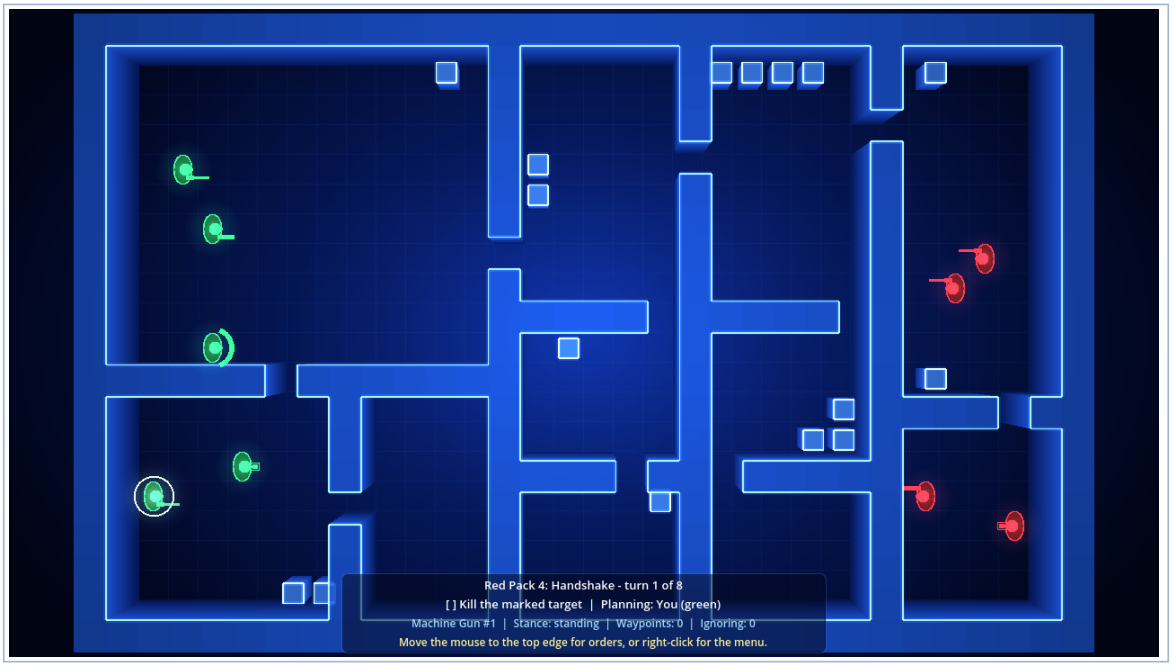
Mission 32 - Hostage Situation. Two hostages (yellow H) held in the middle of an enemy squad; lead both to an edge.



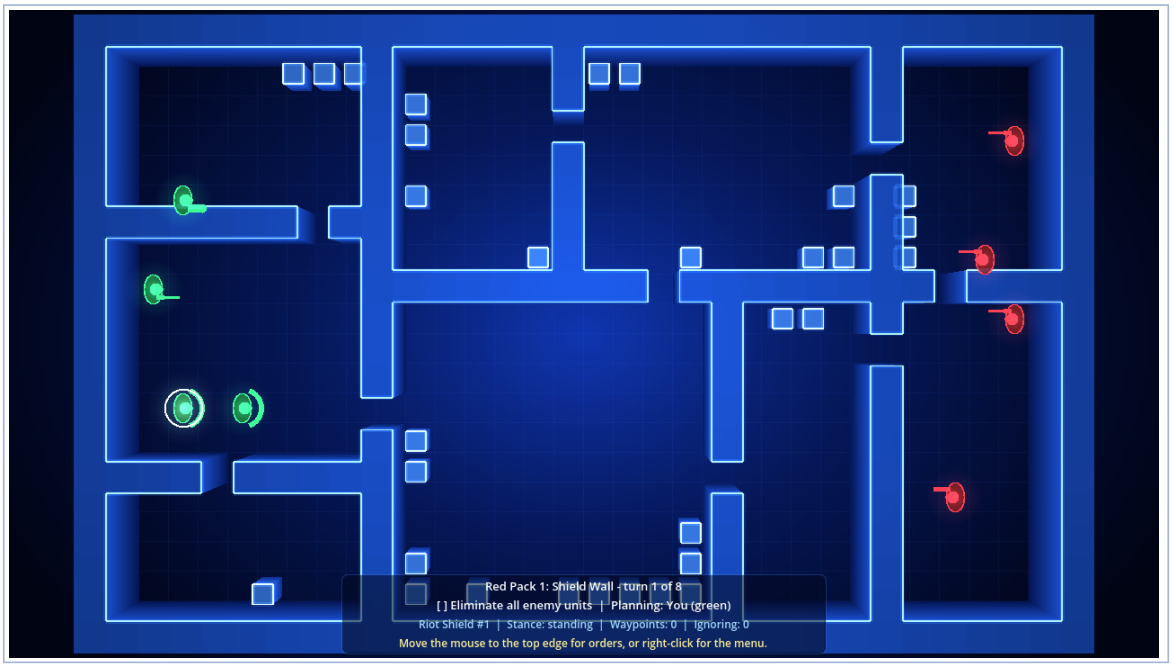
Mission 45 - Protection. One 50-second turn. Three units hold the yellow room while enemies arrive in pairs.



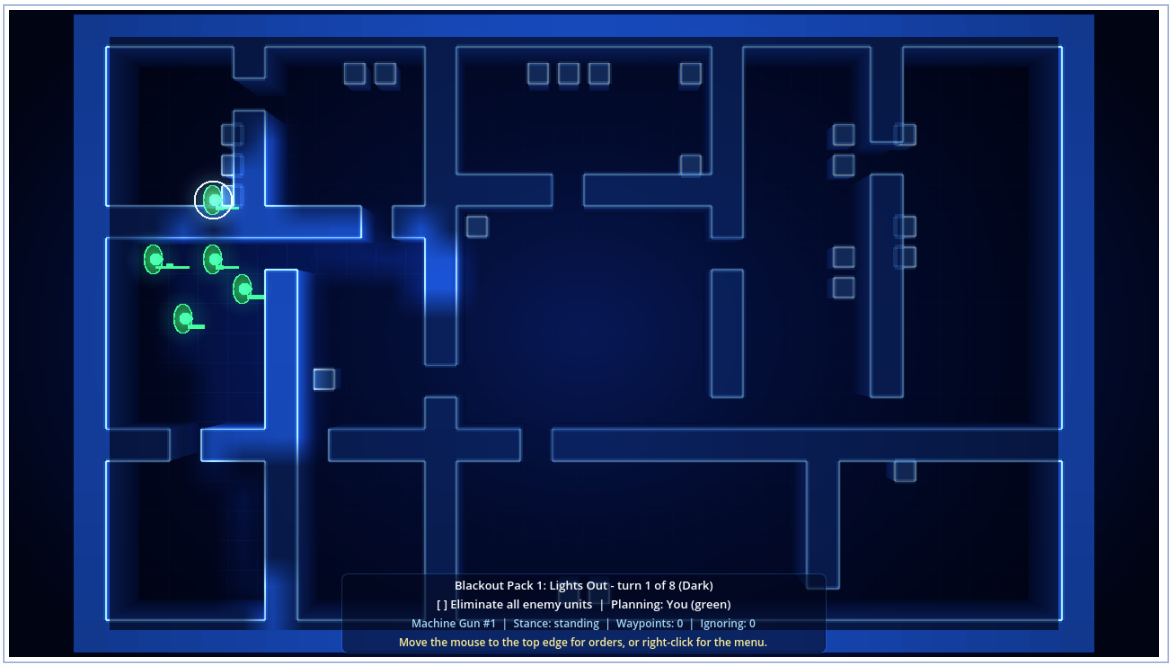
Mission 52 - The Corridor. The Devastator is the VIP and must reach the objective; the enemy defends and does not advance.



Red pack 4 - Handshake. Keep your hacker alive and kill theirs.



Red pack 1 - Shield Wall. Riot shields lead, a grenade launcher works behind them.



Blackout pack 1 - Lights Out. A dark-mode clearance mission; only your cones are lit.

6. Honest limits, and what happens next

Known limits

- **Not play-tested.** Mission difficulty, squad mixes and turn limits are estimates from written hints. Some missions will be too easy or too hard; about half had no hints at all.
- **Maps are generated** (fixed seed per mission), not the original hand-made layouts. Details such as the original's tripwires are not reproduced.
- **Computer opponent:** decent, not strong. In a test against a passive opponent it won 4 and lost 2 of 6; it plans by trying 14 random plans and picking the best.
- **Secure bidding** is not interactive (the zone is chosen by the map generator); Charge bids are fixed in the menu; hostage orders have no dedicated screen.
- **Not yet an installable app:** today it runs from the project folder in Godot 4.3. Packaging is step 5.
- **Interface polish:** no sound and no minimap yet; unit figures are weapon silhouettes only, with no animation. The look is drawn in code, so a true 3D scene or hand-made sprites would be a bigger later step.

How to try it before packaging

Install Godot 4.3 (standard build, Linux x86_64), clone the repository branch `claude/happy-dirac-6dy4yh`, and run `godot --path .` in the project folder. Run all tests with `tests/run_all.sh /path/to/godot`.

Your options

Option	What I would do	Good if
A. Go: package now	Build an AppImage and .deb that include the campaign files, add an app icon and menu entry.	You want the installable app first and will tune balance from real play afterwards.
B. Change first	Tell me what to adjust in this overview: modes, controls, look, mission list, difficulty, names.	Something on these pages is not what you imagined.
C. Tune + polish first	Interactive Secure/Charge bidding, hostage ordering screen, sound effects, unit animations, minimap.	You care more about feel than about installing.

Reply with 'go' for option A, or list changes and I will make them before packaging.